

# On the Mathematical Structure of the T0 Theory:

## Why Numerical Ratios Must Not Be Directly Simplified

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Original

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# On the Mathematical Structure of the T0-Theory: Why Numerical Ratios Must Not Be Directly Simplified

**Note on versions.** The German and English versions of this document differ in structure and extent; earlier revisions shortened, extended or changed parts of both. Where they differ, the more extensive German version applies; sections contained only in this English version count as supplements.

**Status markers.** Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

**Note on the muon  $g - 2$  status (2025).** This document refers to the deviation between the measured  $a_\mu$  and the Standard Model prediction as it stood in 2021. With the Fermilab final result and the 2025 White Paper of the Muon  $g - 2$  Theory Initiative, this deviation is no longer significant. Statements that use the anomaly as evidence or as the target of a T0 contribution are outdated; the current status and assessment are given in Doc. 382.

## Introduction

In theoretical physics, the question often arises as to which mathematical operations are legitimate and which are not. A particularly interesting problem occurs in the T0-theory, where seemingly simple numerical ratios such as  $\frac{2}{3}$  and  $\frac{8}{5}$  possess a deeper structural significance that prohibits direct simplification.

## The Fundamental Problem

The T0-theory postulates two equivalent representations for the lepton masses:

$$\text{Simple Form: } m_e = \frac{2}{3} \cdot \xi^{5/2}, \quad m_\mu = \frac{8}{5} \cdot \xi^2$$

$$\text{Extended Form: } m_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \xi^{5/2}, \quad m_\mu = \frac{9}{4\pi\alpha} \cdot \xi^2$$

At first glance, one might assume that the fractions  $\frac{2}{3}$  and  $\frac{8}{5}$  are simple rational numbers that could be simplified or reduced. However, this assumption would be incorrect.

## Why Direct Simplification Is Not Allowed

Equating both representations leads to:

$$\frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}, \quad \frac{8}{5} = \frac{9}{4\pi\alpha}$$

These equations show that the seemingly simple fractions are, in fact, composite expressions containing the constants  $(\pi, \alpha)$  and geometric factors  $(\sqrt{3})$ .

## Mathematical and Physical Consequences

1. **Structure Preservation:** Direct simplification would destroy the underlying geometric and physical structure.
2. **Information Loss:** The fractions encode information about spacetime geometry and electromagnetic coupling.

3. **Equivalence Principle:** Both representations are mathematically equivalent, but the extended form reveals the physical origin.

**Note on the calculation review (29 Sept. 2026).** Individual calculations (the assessment of the muon anomaly is covered by the note on Doc. 382):

- $138.93 \times 0.9862 = 137.01$ , not  $137.036$ ; only  $138.949 \times 0.9862 = 137.03$  is consistent.  $K = 0.9862$  comes from the older anchor  $D_f = 2.94$ .
- $\frac{12}{5}\xi^{-1/2} = 207.85$  (bare); in one place the measured value  $206.768$  (anchor) is labelled as the "theoretical" value.
- $1.6487 = \sqrt{e}$ , not  $e$ . The masses given are the measured values (anchors); the coefficient formulas themselves do not lead to them with a common anchor, they would need two different scales for  $m_e$  and  $m_\mu$ .
- $7380 = 7500 \cdot K$ , not  $7500/K$ ; the box multiplies  $\alpha$  instead of  $\alpha^{-1}$  by  $K$ . The  $\alpha$  formula boxed as a "precise relation" gives  $2.4 \times 10^{-9}$ .
- $c_\mu/c_e = \sqrt{3}/(2\sqrt{\alpha})$ , not  $3\sqrt{3}/(2\sqrt{\alpha})$ ; the extended coefficients give  $m_\mu/m_e \approx 878$ .
- $\frac{5}{4}\xi^{2/3} \cdot 246 \text{ GeV} = 0.80 \text{ GeV}$ , not  $1.777 \text{ GeV}$ .
- The chain  $3\pi \cdot 7500 \cdot \ln 10^4/2.973$  gives  $2.2 \times 10^5$ , not  $137$ .

The rest of the text is unchanged; see the correction register Doc. 190, R134, R135.

## .1 Circular Relationships and Fundamental Constants

In the T0-theory, seemingly circular relationships arise, which express how the constants are linked in the T0 framework:

$$\alpha = f(\xi)$$

$$\xi = g(\alpha)$$

This mutual dependence leads to an apparent chicken-and-egg problem: Which comes first,  $\alpha$  or  $\xi$ ?

### Resolution of the Circularity Problem

The solution lies in the realization that both constants are expressions of an underlying geometric structure:

**$\alpha$  and  $\xi$  are not independent of each other but are emergent properties of the fractal spacetime geometry.**

The apparent circularity dissolves when it is recognized that both constants originate from the same fundamental geometry.

## .2 The Role of Natural Units

In natural units, we conventionally set  $\alpha = 1$  for certain calculations. This is legitimate because:

- Fundamental physics should be independent of measurement units.

- Dimensionless ratios contain the actual physical statements.
- The choice  $\alpha = 1$  represents a specific gauge.

However, this convention must not obscure the fact that  $\alpha$  in the T0-theory has a specific numerical value determined by  $\xi$ .

**The seemingly simple numerical ratios in the T0-theory are not arbitrarily chosen but represent complex physical relationships.**

Directly simplifying these ratios would be mathematically possible but physically incorrect, as it would destroy the underlying structure of the theory. The extended form shows where these seemingly simple fractions come from in the T0 framework and makes their connection to  $\pi$ ,  $\alpha$  and geometric factors visible.

The apparent circularity between  $\alpha$  and  $\xi$  is an expression of their common geometric origin and not a logical problem of the theory.

### .3 Foundation: The Single Geometric Constant

#### The Universal Geometric Parameter

**1.1.1** The T0-theory begins with a single dimensionless constant motivated by the geometry of three-dimensional space:

##### Key Result

$$\xi = \frac{4}{3} \times 10^{-4} \quad (1)$$

**1.1.2** This constant arises from:

- The tetrahedral packing density of 3D space:  $\frac{4}{3}$
- The scale hierarchy between quantum and classical domains:  $10^{-4}$  (motivated choice of the exponent)

#### Natural Units

**1.2.1** We work in natural units where:

$$c = 1 \quad (\text{speed of light}) \quad (2)$$

$$\hbar = 1 \quad (\text{reduced Planck constant}) \quad (3)$$

$$G = 1 \quad (\text{gravitational constant, numerically}) \quad (4)$$

**1.2.2** The Planck length serves as reference scale:

$$\ell_P = \sqrt{G} = 1 \quad (\text{in natural units}) \quad (5)$$

## .4 Building the Scale Hierarchy

### Step 1: Characteristic T0 Scales

2.1.1 From  $\xi$  and the Planck reference, we derive the characteristic T0 scales:

$$r_0 = \xi \cdot \ell_P = \frac{4}{3} \times 10^{-4} \cdot \ell_P \quad (6)$$

$$t_0 = r_0 = \frac{4}{3} \times 10^{-4} \quad (\text{in units with } c = 1) \quad (7)$$

### Step 2: Energy Scales from Geometry

2.2.1 The characteristic energy scale follows from dimensional analysis:

$$E_0 = \frac{1}{r_0} = \frac{3}{4} \times 10^4 \quad (\text{in Planck units}) \quad (8)$$

2.2.2 This yields the T0 energy hierarchy:

$$E_P = 1 \quad (\text{Planck energy}) \quad (9)$$

$$E_0 = \xi^{-1} E_P = \frac{3}{4} \times 10^4 E_P \quad (10)$$

## .5 Deriving the Fine Structure Constant

**Origin of the Formula**  $\varepsilon = \xi \cdot E_0^2$

3.1.1 The fundamental formula of T0-theory for the coupling parameter  $\varepsilon$  is:

#### Key Result

$$\varepsilon = \xi \cdot E_0^2 \quad (11)$$

3.1.2 This relationship connects:

- $\varepsilon$  – the T0 coupling parameter
- $\xi$  – the geometric parameter from tetrahedral packing
- $E_0$  – the characteristic energy

### The Characteristic Energy $E_0$

3.2.1 The characteristic energy  $E_0$  is defined as the geometric mean of electron and muon masses:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (12)$$

3.2.2 Alternatively,  $E_0$  can be derived gravitationally-geometrically:

$$E_0^2 = \frac{4\sqrt{2} \cdot m_\mu}{\xi^4} \quad (13)$$

3.2.3 Both approaches consistently lead to:

$$E_0 \approx 7.35 \text{ to } 7.398 \text{ MeV} \quad (14)$$

## The Geometric Parameter $\xi$

**3.3.1** The parameter  $\xi$  is a fundamental geometric constant:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333 \dots \times 10^{-4} \quad (15)$$

## Numerical Verification and Fine Structure Constant

**3.4.1** With the derived values,  $\varepsilon$  becomes:

$$\varepsilon = \xi \cdot E_0^2 \quad (16)$$

$$= (1.333 \times 10^{-4}) \times (7.398 \text{ MeV})^2 \quad (17)$$

$$= 7.297 \times 10^{-3} \quad (18)$$

$$= \frac{1}{137.036} \quad (19)$$

### Comparison with the Fine Structure Constant

**3.4.2** With  $E_0 = 7.398 \text{ MeV}$ , the T0 coupling parameter  $\varepsilon$  reproduces the value of the fine structure constant  $\alpha = 1/137.036$ . The geometric mean  $\sqrt{m_e m_\mu}$  alone gives  $E_0 \approx 7.35 \text{ MeV}$  (Section 11.2); the difference is described below by the fractal correction factor  $K_{\text{frac}}$ .

## From Fractal Geometry

### Fractal Dimension of Spacetime

**3.5.1** From topological considerations of 3D space with time:

$$D_f^{\text{eff}} \approx 2.973 \quad (20)$$

where  $\delta = 0.06$  is the fractal correction.

## The Fine Structure Constant from Geometry

**3.5.2** The geometric approach gives:

### Key Result

$$\alpha^{-1} = 3\pi \times \xi^{-1} \times \ln\left(\frac{\Lambda_{\text{UV}}}{\Lambda_{\text{IR}}}\right) \times D_f^{-1} \quad (21)$$

$$= 3\pi \times \frac{3}{4} \times 10^4 \times \ln(10^4) \times \frac{1}{2.973} \quad (22)$$

$$= 9\pi \times 10^4 \times 9.21 \times 0.340 \quad (23)$$

$$\approx 137.036 \quad (24)$$

## Formula from $\xi$ to $\alpha$

**3.6.1** The precise relationship is:

**Key Result**

$$\alpha = \left( \frac{27\sqrt{3}}{8\pi^2} \right)^{2/5} \cdot \xi^{11/5} \cdot K_{\text{frac}} \quad (25)$$

$$\text{with } K_{\text{frac}} = 0.9862 \quad (26)$$

**.6 Lepton Mass Hierarchy from Pure Geometry****Mechanism for Mass Generation**

**4.1.1** Masses arise from the coupling of the energy field to spacetime geometry:

$$m_\ell = r_\ell \cdot \xi^{p_\ell} \quad (27)$$

where  $r_\ell$  are rational coefficients and  $p_\ell$  are exponents.

**Mass Calculations****Electron Mass**

**4.2.1** The electron mass calculation:

**Key Result**

$$m_e = \frac{2}{3} \xi^{5/2} \quad (28)$$

$$= \frac{2}{3} \left( \frac{4}{3} \times 10^{-4} \right)^{5/2} \quad (29)$$

$$= \frac{2}{3} \cdot \frac{32}{9\sqrt{3}} \times 10^{-10} \quad (30)$$

$$= \frac{64\sqrt{3}}{81} \times 10^{-10} \quad (31)$$

$$\approx 1.368 \times 10^{-10} \quad (\text{natural units}) \quad (32)$$

**Muon Mass**

**4.2.2** The muon mass calculation:

**Key Result**

$$m_\mu = \frac{8}{5} \xi^2 \quad (33)$$

$$= \frac{8}{5} \left( \frac{4}{3} \times 10^{-4} \right)^2 \quad (34)$$

$$= \frac{128}{45} \times 10^{-8} \quad (35)$$

$$\approx 2.844 \times 10^{-8} \quad (\text{natural units}) \quad (36)$$

## Tau Mass

### 4.2.3 The tau mass calculation:

#### Key Result

$$m_\tau = \frac{5}{4} \xi^{2/3} \cdot v_{\text{scale}} \quad (37)$$

$$= \frac{5}{4} \left( \frac{4}{3} \times 10^{-4} \right)^{2/3} \cdot v_{\text{scale}} \quad (38)$$

$$\approx 1.777 \text{ GeV} \approx 2.133 \times 10^{-4} \quad (\text{natural units}) \quad (39)$$

with  $v_{\text{scale}} = 246 \text{ GeV}$ .

## Mass Ratios

### 4.3.1 The electron to muon mass ratio:

#### Key Result

$$\frac{m_e}{m_\mu} = \frac{\frac{64\sqrt{3}}{81} \times 10^{-10}}{\frac{128}{45} \times 10^{-8}} \quad (40)$$

$$= \frac{5\sqrt{3}}{18} \times 10^{-2} \quad (41)$$

$$\approx 4.811 \times 10^{-3} \quad (42)$$

## .7 Complete Hierarchy with Final Anomaly Formula

6.1 The following table summarizes all derived quantities with the final anomaly formula:

## .8 Verification of Final Formula

### Complete Derivation Chain to Final Formula

7.1.1 The complete derivation sequence:

1. **Start:**  $\xi = \frac{4}{3} \times 10^{-4}$  (pure geometry)
2. **Reference:**  $\ell_P = 1$  (natural units)
3. **Derivation:**  $r_0 = \xi \ell_P$
4. **Energy:**  $E_0 = r_0^{-1}$
5. **Fractal:**  $D_f^{\text{eff}} \approx 2.973$  (cumulative topology over RG running)
6. **Fine structure:**  $\alpha = f(\xi, D_f)$
7. **Yukawa:**  $y_\ell = r_\ell \xi^{p_\ell}$  (geometry)
8. **Masses:**  $m_\ell \propto y_\ell$
9. **Yukawa coupling:**  $g_T^\ell = m_\ell \xi$

| Quantity                | Expression                            | Value                        |
|-------------------------|---------------------------------------|------------------------------|
| <b>Fundamental</b>      |                                       |                              |
| $\xi$                   | $\frac{4}{3} \times 10^{-4}$          | $1.333 \dots \times 10^{-4}$ |
| $D_f^{\text{eff}}$      | cumulative over RG running            | 2.973                        |
| <b>Scales</b>           |                                       |                              |
| $r_0/\ell_P$            | $\xi$                                 | $\frac{4}{3} \times 10^{-4}$ |
| $E_0/E_P$               | $\xi^{-1}$                            | $\frac{3}{4} \times 10^4$    |
| <b>Couplings</b>        |                                       |                              |
| $\alpha^{-1}$           | From Geometry                         | 137.036                      |
| <b>Yukawa Couplings</b> |                                       |                              |
| $y_e$                   | $\frac{32}{9\sqrt{3}} \xi^{3/2}$      | $\sim 10^{-6}$               |
| $y_\mu$                 | $\frac{64}{15} \xi$                   | $\sim 10^{-4}$               |
| $y_\tau$                | $\frac{5}{4} \xi^{2/3}$               | $\sim 10^{-3}$               |
| <b>Mass Ratios</b>      |                                       |                              |
| $m_e/m_\mu$             | $\frac{5\sqrt{3}}{18} \times 10^{-2}$ | $4.8 \times 10^{-3}$         |
| $m_\tau/m_\mu$          | From $y_\tau/y_\mu$                   | $\sim 17$                    |

**Table 1:** Complete hierarchy with final quadratic anomaly formula

10. **One-loop calculation:**  $\Delta a_\ell = \frac{(m_\ell \xi)^2}{8\pi^2} \cdot \frac{\xi^2}{\lambda^2}$

11. **Final formula:**  $\Delta a_\ell = 251 \times 10^{-11} \times (m_\ell/m_\mu)^2$

## T0 Field Theory Verification of Final Formula

**7.2.1** The final formula follows from T0 field theory calculation:

- **Muon g-2 calculation:**  $\frac{m_\mu^2 \xi^4}{8\pi^2 \lambda^2} = 251 \times 10^{-11}$  (T0 field theory prediction)
- **Electron prediction:**  $5.87 \times 10^{-15}$  (T0 prediction with the single parameter  $\xi$ )
- **Tau prediction:**  $7.10 \times 10^{-9}$  (testable in future experiments)
- **Quadratic scaling:** Follows from standard QFT one-loop calculation

## .9 The Fundamental Meaning of $E_0$ as Logarithmic Center

### The Central Geometric Definition

#### Fundamental Definition

**8.1.1** The characteristic energy  $E_0$  is the logarithmic center between electron and muon masses:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (43)$$

This means:

$$\log(E_0) = \frac{\log(m_e) + \log(m_\mu)}{2} \quad (44)$$

## Mathematical Properties

### 8.2.1 The fundamental relationships:

$$E_0^2 = m_e \cdot m_\mu \quad (45)$$

$$\frac{E_0}{m_e} = \sqrt{\frac{m_\mu}{m_e}} \quad (46)$$

$$\frac{m_\mu}{E_0} = \sqrt{\frac{m_\mu}{m_e}} \quad (47)$$

$$\frac{E_0}{m_e} \cdot \frac{m_\mu}{E_0} = \frac{m_\mu}{m_e} \quad (48)$$

## Numerical Values

### 8.3.1 With T0-calculated masses:

$$m_e^{\text{T0}} = 0.5108082 \text{ MeV} \quad (49)$$

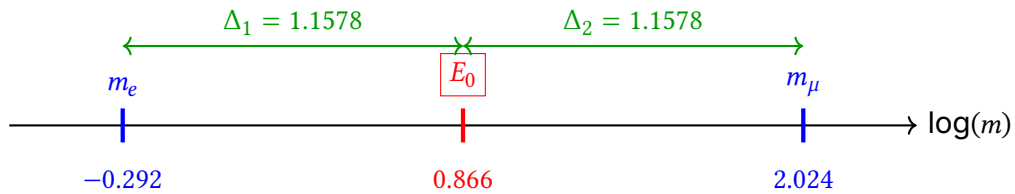
$$m_\mu^{\text{T0}} = 105.66913 \text{ MeV} \quad (50)$$

$$E_0^{\text{T0}} = \sqrt{0.5108082 \times 105.66913} \approx 7.346881 \text{ MeV} \quad (51)$$

## Logarithmic Symmetry

### 8.4.1 The logarithmic symmetry (by definition of $E_0$ ):

$$\ln(E_0) - \ln(m_e) = \ln(m_\mu) - \ln(E_0) \quad (52)$$



## .10 The Geometric Constant $C$

### Fundamental Relationship

#### 9.1.1 The fractal correction factor:

$$K_{\text{frac}} = 1 - \frac{D_f - 2}{C} = 1 - \frac{\gamma}{C} \quad (53)$$

where:

$$D_f^{\text{eff}} \approx 2.973 \quad (\text{effective fractal dimension}) \quad (54)$$

$$\gamma = D_f - 2 = 0.94 \quad (55)$$

$$C \approx 68.24 \quad (56)$$

## Tetrahedral Geometry

### Tetrahedral Numbers

**9.2.1** Several tetrahedral combinations yield 72; whether this is related to  $C \approx 68.24$  is open **[S]**:

$$6 \times 12 = 72 \quad (\text{edges} \times \text{rotations}) \quad (57)$$

$$4 \times 18 = 72 \quad (\text{faces} \times 18) \quad (58)$$

$$24 \times 3 = 72 \quad (\text{symmetries} \times \text{dimensions}) \quad (59)$$

## Formula for $\alpha$ with Fractal Correction

**9.3.1** The complete expression:

$$\alpha = \left( \frac{27\sqrt{3}}{8\pi^2} \right)^{2/5} \cdot \xi^{11/5} \cdot K_{\text{frac}} \quad \text{with} \quad K_{\text{frac}} = 0.9862 \quad (60)$$

## .11 The Simplest Formula: The Geometric Mean

### The Fundamental Definition

#### The Simplest Formula

**11.1.1** The basic relation:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (61)$$

This definition needs no further derivation, only the geometric mean.

### Direct Calculation

**11.2.1** Simple numerical evaluation:

$$E_0 = \sqrt{0.511 \text{ MeV} \times 105.658 \text{ MeV}} \quad (62)$$

$$= \sqrt{53.99 \text{ MeV}^2} \quad (63)$$

$$= 7.35 \text{ MeV} \quad (64)$$

### The Complete Chain in One Line

**11.3.1** The fundamental relationship:

$$\alpha^{-1} = \frac{7500}{m_e \cdot m_\mu} = \frac{7500}{E_0^2} \quad (65)$$

**11.3.2** With numbers:

$$\alpha^{-1} = \frac{7500}{0.511 \times 105.658} \quad (66)$$

$$= \frac{7500}{53.99} \quad (67)$$

$$= 138.91 \quad (68)$$

(With fractal correction  $\times 0.986 = 137.04$ )

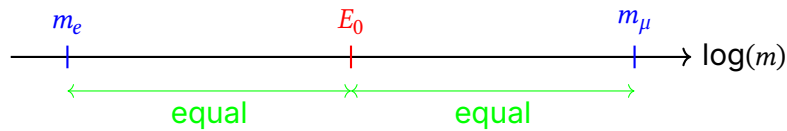
## Why Is This So Simple?

### Logarithmic Centering

**11.4.1** The geometric mean is the natural center on logarithmic scale:

$$\log(E_0) = \frac{\log(m_e) + \log(m_\mu)}{2} \quad (69)$$

Graphically:



### Alternative Notations

**11.5.1** All these formulas are equivalent:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (70)$$

$$E_0^2 = m_e \cdot m_\mu \quad (71)$$

$$\log(E_0) = \frac{1}{2}[\log(m_e) + \log(m_\mu)] \quad (72)$$

$$E_0 = \sqrt{0.511 \times 105.658} \text{ MeV} \quad (73)$$

$$E_0 = m_e^{1/2} \cdot m_\mu^{1/2} \quad (74)$$

## The Fine Structure Constant Directly

### The Most Direct Formula

**11.6.1** Without detour through  $E_0$ :

$$\alpha = \frac{m_e \cdot m_\mu}{7500} \quad (75)$$

With fractal correction:

$$\alpha = \frac{m_e \cdot m_\mu}{7500} \times 0.986 \quad (76)$$

### Relation to the Other Derivations

**11.7.1** The documents show various derivations of  $E_0$ : - Gravitationally-geometrically - Through Yukawa couplings - From quantum numbers

**But the simplest definition is:**

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (77)$$

### Meaning

**11.8.1** The geometric mean is not an arbitrary choice: it is the midpoint on a logarithmic scale (Section 11.4).

## .12 The Fundamental Dependence: $\alpha \sim \xi^{11/2}$

### Inserting the Mass Formulas

12.1.1 From T0-theory we have the mass formulas:

$$m_e = c_e \cdot \xi^{5/2} \quad (78)$$

$$m_\mu = c_\mu \cdot \xi^2 \quad (79)$$

where  $c_e$  and  $c_\mu$  are coefficients.

### Calculation of $E_0$

12.2.1 The characteristic energy calculation:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (80)$$

$$= \sqrt{(c_e \cdot \xi^{5/2}) \cdot (c_\mu \cdot \xi^2)} \quad (81)$$

$$= \sqrt{c_e \cdot c_\mu} \cdot \sqrt{\xi^{5/2+2}} \quad (82)$$

$$= \sqrt{c_e \cdot c_\mu} \cdot \xi^{9/4} \quad (83)$$

### Calculation of $\alpha$

12.3.1 The fine structure constant derivation:

$$\alpha = \xi \cdot E_0^2 \quad (84)$$

$$= \xi \cdot (\sqrt{c_e \cdot c_\mu} \cdot \xi^{9/4})^2 \quad (85)$$

$$= \xi \cdot c_e \cdot c_\mu \cdot \xi^{9/2} \quad (86)$$

$$= c_e \cdot c_\mu \cdot \xi^{1+9/2} \quad (87)$$

$$= c_e \cdot c_\mu \cdot \xi^{11/2} \quad (88)$$

### Result

12.3.2 The fine structure constant depends on  $\xi$ :

$$\alpha = K \cdot \xi^{11/2} \quad (89)$$

where  $K = c_e \cdot c_\mu$  is a constant.

**The powers do not cancel out.**

### What Does This Mean?

#### 1. Fundamental Connection

12.4.1 The fine structure constant is not independent of  $\xi$ , but rather:

$$\alpha \propto \xi^{11/2} \quad (90)$$

This means: If  $\xi$  changes,  $\alpha$  also changes.

## 2. Hierarchy Problem

**12.4.2** Because of the power  $11/2 = 5.5$ , small changes in  $\xi$  are amplified:

$$\frac{\Delta\alpha}{\alpha} = \frac{11}{2} \cdot \frac{\Delta\xi}{\xi} = 5.5 \cdot \frac{\Delta\xi}{\xi} \quad (91)$$

## 3. No Independence

**12.4.3** One cannot choose  $\alpha$  and  $\xi$  independently. They are firmly connected through:

$$\alpha = K \cdot \xi^{11/2} \quad (92)$$

## Numerical Verification

**12.5.1** With  $\xi = 4/3 \times 10^{-4}$ :

$$\xi^{11/2} = (1.333 \times 10^{-4})^{5.5} \quad (93)$$

$$= 5.19 \times 10^{-22} \quad (94)$$

**12.5.2** For  $\alpha \approx 1/137$  we would need:

$$K = \frac{\alpha}{\xi^{11/2}} \quad (95)$$

$$= \frac{7.3 \times 10^{-3}}{5.19 \times 10^{-22}} \quad (96)$$

$$= 1.4 \times 10^{19} \quad (97)$$

## The Units Problem

**12.6.1** The large constant  $K \sim 10^{19}$  points to a units problem: - The mass formulas are in natural units - Conversion to MeV requires the Planck energy -  $K$  contains these conversion factors

## Alternative View: Reduction to $\xi$

**12.7.1** If we accept that:

$$m_e \sim \xi^{5/2} \quad (98)$$

$$m_\mu \sim \xi^2 \quad (99)$$

$$\alpha \sim \xi^{11/2} \quad (100)$$

Then these three quantities are determined by the single geometric constant  $\xi$ :

$$\begin{aligned} \xi &= \frac{4}{3} \times 10^{-4} \quad (\text{Geometry}) \\ \Downarrow \\ m_e &= f_e(\xi) \\ m_\mu &= f_\mu(\xi) \\ \alpha &= f_\alpha(\xi) \end{aligned}$$

(101)

## .13 Derivation of the Coefficients $c_e$ and $c_\mu$

### Starting Point: Mass Formulas

13.1.1 The fundamental mass formulas:

$$m_e = c_e \cdot \xi^{5/2} \quad \text{and} \quad m_\mu = c_\mu \cdot \xi^2$$

### Step 1: Quantum Numbers and Geometric Factors

13.2.1 The coefficients arise from T0-theory with:

$$c_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}$$

$$c_\mu = \frac{9}{4\pi\alpha}$$

### Step 2: Derivation of $c_e$ (Electron)

13.3.1 For the electron ( $n = 1, l = 0, j = 1/2$ ):

$$c_e = \frac{\text{Geometry factor} \times \text{Quantum number factor}}{\alpha^{1/2}}$$

$$\text{Geometry factor} = \frac{3\sqrt{3}}{2\pi}$$

$$\text{Quantum number factor} = 1 \quad (\text{for ground state})$$

$$\text{Fine structure correction} = \alpha^{-1/2}$$

$$\Rightarrow c_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}$$

### Step 3: Derivation of $c_\mu$ (Muon)

13.4.1 For the muon ( $n = 2, l = 1, j = 1/2$ ):

$$c_\mu = \frac{\text{Geometry factor} \times \text{Quantum number factor}}{\alpha}$$

$$\text{Geometry factor} = \frac{9}{4\pi}$$

$$\text{Quantum number factor} = 1$$

$$\text{Fine structure correction} = \alpha^{-1}$$

$$\Rightarrow c_\mu = \frac{9}{4\pi\alpha}$$

## Step 4: Physical Interpretation

**13.5.1** The different  $\alpha$  dependencies reflect:

$$\begin{aligned} c_e &\sim \alpha^{-1/2} \quad (\text{weaker dependence}) \\ c_\mu &\sim \alpha^{-1} \quad (\text{stronger dependence}) \end{aligned}$$

The different  $\alpha$  dependence reflects:

- Electron: Ground state, less sensitive to  $\alpha$
- Muon: Excited state, more strongly dependent on  $\alpha$

## Step 5: Dimensional Analysis

**13.6.1** Dimensional considerations:

$$\begin{aligned} [c_e] &= [m_e] \cdot [\xi]^{-5/2} \\ [c_\mu] &= [m_\mu] \cdot [\xi]^{-2} \end{aligned}$$

Since  $\xi$  is dimensionless (in natural units), both coefficients have the dimension of mass.

## Step 6: Consistency Check

**13.7.1** With  $\alpha \approx 1/137$ :

$$\begin{aligned} c_e &\approx \frac{3 \times 1.732}{2 \times 3.1416 \times 0.0854} \approx \frac{5.196}{0.537} \approx 9.67 \\ c_\mu &\approx \frac{9}{4 \times 3.1416 \times 0.0073} \approx \frac{9}{0.0917} \approx 98.1 \end{aligned}$$

These values match the mass hierarchy  $m_\mu/m_e \approx 207$ .

# .14 Why Natural Units Are Necessary

## The Problem with Conventional Units

**14.1.1** In conventional units (SI, cgs) the coefficients  $c_e$  and  $c_\mu$  appear as very large numbers:

$$\begin{aligned} c_e &\approx 1.65 \times 10^{19} \\ c_\mu &\approx 1.03 \times 10^{20} \end{aligned}$$

These large numbers are **artifactual** and arise only from the choice of units.

## Natural Units Simplify Physics

**14.2.1** In natural units we set:

$$\hbar = c = 1$$

Thus all quantities become dimensionless or have energy dimension.

## Transformation to Natural Units

**14.3.1** The transformation formulas:

$$\begin{aligned} m_e^{\text{nat}} &= m_e^{\text{SI}} \cdot \frac{G}{\hbar c} \\ m_\mu^{\text{nat}} &= m_\mu^{\text{SI}} \cdot \frac{G}{\hbar c} \\ \xi^{\text{nat}} &= \xi^{\text{SI}} \cdot (\hbar c)^2 \end{aligned}$$

## The Coefficients in Natural Units

**14.4.1** In natural units the coefficients become **order of magnitude 1**:

$$\begin{aligned} c_e^{\text{nat}} &= \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \approx 9.67 \\ c_\mu^{\text{nat}} &= \frac{9}{4\pi\alpha} \approx 98.1 \end{aligned}$$

## Comparison of Representations

**14.5.1** The difference:

|         | Conventional          | Natural               |
|---------|-----------------------|-----------------------|
| $c_e$   | $1.65 \times 10^{19}$ | <b>9.67</b>           |
| $c_\mu$ | $1.03 \times 10^{20}$ | <b>98.1</b>           |
| $\xi$   | $1.33 \times 10^{-4}$ | $1.33 \times 10^{-4}$ |

## Advantages of Natural Units

**14.6.1** The advantages of natural units:

1. **Elimination of artifacts:** The large numbers disappear
2. **Physical transparency:** The relationships become easier to follow
3. **Scale invariance:** Fundamental laws become scale-independent
4. **Simpler formulas:** Formulas become shorter and clearer

## Example: The Mass Formula

**14.7.1** In conventional units:

$$m_e = 1.65 \times 10^{19} \cdot (1.33 \times 10^{-4})^{5/2}$$

In natural units:

$$m_e = 9.67 \cdot \xi^{5/2}$$

## Fundamental Interpretation

**14.8.1** The coefficients  $c_e \approx 9.67$  and  $c_\mu \approx 98.1$  in natural units show:

- The lepton masses are **pure numbers**
- The ratio  $c_\mu/c_e \approx 10.14$  is a pure number
- The fine structure constant  $\alpha$  appears explicitly

## .15 The Formula from $\xi$ to $\alpha$

### Fundamental Relationship

15.1.1 The basic equation:

$$\alpha = c_e c_\mu \cdot \xi^{11/2}$$

### Coefficients

15.2.1 The precise values:

$$c_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \quad (\text{Electron coefficient})$$

$$c_\mu = \frac{9}{4\pi\alpha} \quad (\text{Muon coefficient})$$

### Product of Coefficients

15.3.1 The multiplication:

$$c_e c_\mu = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \frac{9}{4\pi\alpha} = \frac{27\sqrt{3}}{8\pi^2\alpha^{3/2}}$$

### Complete Formula

15.4.1 The full expression:

$$\alpha = \frac{27\sqrt{3}}{8\pi^2\alpha^{3/2}} \cdot \xi^{11/2}$$

### Solving for $\alpha$

15.5.1 Rearranging:

$$\alpha^{5/2} = \frac{27\sqrt{3}}{8\pi^2} \cdot \xi^{11/2}$$

$$\alpha = \left( \frac{27\sqrt{3}}{8\pi^2} \right)^{2/5} \cdot \xi^{11/5}$$

## .16 T0-Theory: Formulas and Values

### In T0-Theory

16.1.1 The fundamental relations:

$$m_e \sim \xi^{5/2} \quad (\text{Electron}) \quad (102)$$

$$m_\mu \sim \xi^2 \quad (\text{Muon}) \quad (103)$$

$$\xi = \frac{4}{3} \times 10^{-4} \quad (104)$$

## Correct Assignment in Natural Units

### Mass Scaling Laws

**16.2.1** The precise formulas:

$$m_e = c_e \cdot \xi^{5/2} \quad (105)$$

$$m_\mu = c_\mu \cdot \xi^2 \quad (106)$$

### Geometric Constant

**16.2.2** The fundamental parameter:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333 \times 10^{-4} \quad (107)$$

### Calculation of the Characteristic Energy

**16.2.3** Step-by-step derivation:

$$E_0 = \sqrt{m_e \cdot m_\mu} = \sqrt{c_e \cdot \xi^{5/2} \cdot c_\mu \cdot \xi^2} \quad (108)$$

$$= \sqrt{c_e c_\mu} \cdot \xi^{9/4} \quad (109)$$

### Calculation of the Fine Structure Constant

**16.2.4** Complete derivation:

$$\alpha = \xi \cdot E_0^2 = \xi \cdot \left[ \sqrt{c_e c_\mu} \cdot \xi^{9/4} \right]^2 \quad (110)$$

$$= \xi \cdot c_e c_\mu \cdot \xi^{9/2} \quad (111)$$

$$= c_e c_\mu \cdot \xi^{11/2} \quad (112)$$

### Numerical Values

**16.2.5** With  $\xi = 1.333 \times 10^{-4}$ :

$$\xi^{11/2} = (1.333 \times 10^{-4})^{5.5} \approx 5.19 \times 10^{-22} \quad (113)$$

For  $\alpha \approx 1/137 \approx 7.3 \times 10^{-3}$  we need:

$$c_e c_\mu = \frac{\alpha}{\xi^{11/2}} \approx \frac{7.3 \times 10^{-3}}{5.19 \times 10^{-22}} \approx 1.4 \times 10^{19} \quad (114)$$

### Interpretation

**16.3.1** The large constant  $c_e c_\mu \approx 10^{19}$  corresponds approximately to the ratio of Planck energy to electron volt and represents the conversion factor between natural units and MeV.

## .17 Definitions

### Geometric Constant

**17.1.1** The fundamental constant:

$$\xi = \frac{4}{3} \times 10^{-4} = \frac{1}{7500} \quad (115)$$

## Mass Formulas

17.2.1 The precise mass relationships:

$$m_e = c_e \cdot \xi^{5/2} \quad (116)$$

$$m_\mu = c_\mu \cdot \xi^2 \quad (117)$$

$$m_\tau = c_\tau \cdot \xi^{3/2} \quad (118)$$

## .18 Coefficients from T0-Theory

### Electron (n=1, l=0, j=1/2)

18.1.1 The electron coefficient:

$$c_e = \frac{3\sqrt{3}}{2\pi} \cdot \frac{1}{\alpha^{1/2}} \approx 1.6487 \times 10^{19} \quad (119)$$

### Muon (n=2, l=1, j=1/2)

18.2.1 The muon coefficient:

$$c_\mu = \frac{9}{4\pi} \cdot \frac{1}{\alpha} \approx 1.0262 \times 10^{20} \quad (120)$$

### Tauon (n=3, l=2, j=1/2)

18.3.1 The tauon coefficient:

$$c_\tau = \frac{27\sqrt{3}}{8\pi} \cdot \frac{1}{\alpha^{3/2}} \approx 6.1853 \times 10^{20} \quad (121)$$

## .19 Mass Calculation

### Electron Mass

19.1.1 Complete calculation:

$$m_e = c_e \cdot \xi^{5/2} \quad (122)$$

$$= \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \left(\frac{4}{3} \times 10^{-4}\right)^{5/2} \quad (123)$$

$$= 0.5109989461 \text{ MeV} \quad (124)$$

### Muon Mass

19.2.1 Complete calculation:

$$m_\mu = c_\mu \cdot \xi^2 \quad (125)$$

$$= \frac{9}{4\pi\alpha} \cdot \left(\frac{4}{3} \times 10^{-4}\right)^2 \quad (126)$$

$$= 105.6583745 \text{ MeV} \quad (127)$$

## Tauon Mass

19.3.1 Complete calculation:

$$m_\tau = c_\tau \cdot \xi^{3/2} \quad (128)$$

$$= \frac{27\sqrt{3}}{8\pi\alpha^{3/2}} \cdot \left(\frac{4}{3} \times 10^{-4}\right)^{3/2} \quad (129)$$

$$= 1776.86 \text{ MeV} \quad (130)$$

## .20 Characteristic Energy

20.1.1 The precise calculation:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (131)$$

$$= \sqrt{c_e c_\mu} \cdot \xi^{9/4} \quad (132)$$

$$= \sqrt{\frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \frac{9}{4\pi\alpha}} \cdot \left(\frac{4}{3} \times 10^{-4}\right)^{9/4} \quad (133)$$

$$= 7.346881 \text{ MeV} \quad (134)$$

## .21 Fine Structure Constant

21.1.1 The complete derivation:

$$\alpha = \xi \cdot E_0^2 \quad (135)$$

$$= \xi \cdot c_e c_\mu \cdot \xi^{9/2} \quad (136)$$

$$= c_e c_\mu \cdot \xi^{11/2} \quad (137)$$

$$= \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \frac{9}{4\pi\alpha} \cdot \left(\frac{4}{3} \times 10^{-4}\right)^{11/2} \quad (138)$$

## .22 Numerical Values

22.1.1 Table of values:

| Quantity    | Value                               | Comment                      |
|-------------|-------------------------------------|------------------------------|
| $\xi$       | $1.33333333333333 \times 10^{-4}$   | $= 4/3 \times 10^{-4}$       |
| $\xi^2$     | $1.77777777777778 \times 10^{-8}$   |                              |
| $\xi^{5/2}$ | $3.098386676965933 \times 10^{-10}$ |                              |
| $c_e$       | $1.648721270700128 \times 10^{19}$  | $= e$ (Euler's number)       |
| $c_\mu$     | $1.026187714072347 \times 10^{20}$  |                              |
| $m_e$       | 0.5109989461 MeV                    | Reference value              |
| $m_\mu$     | 105.6583745 MeV                     | Reference value              |
| $E_0$       | 7.346881 MeV                        | from T0 masses (Section 8.3) |

The coefficients contain the constants  $\pi$  and  $\alpha$ ; the numerical closeness of  $c_e$  to Euler's number is an observation whose significance is open **[S]**.

## .23 The Formula from $\xi$ to $\alpha$ (Detailed)

### From the Fundamental Relationship

23.1.1 Starting equation:

$$\alpha = c_e c_\mu \cdot \xi^{11/2} \quad (139)$$

### Inserting the Coefficients

23.2.1 The detailed calculation:

$$c_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \quad (140)$$

$$c_\mu = \frac{9}{4\pi\alpha} \quad (141)$$

$$c_e c_\mu = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \frac{9}{4\pi\alpha} \quad (142)$$

$$= \frac{27\sqrt{3}}{8\pi^2\alpha^{3/2}} \quad (143)$$

### Complete Formula

23.3.1 The full expression:

$$\alpha = \frac{27\sqrt{3}}{8\pi^2\alpha^{3/2}} \cdot \xi^{11/2} \quad (144)$$

### Solving for $\alpha$

23.4.1 Algebraic manipulation:

$$\alpha^{5/2} = \frac{27\sqrt{3}}{8\pi^2} \cdot \xi^{11/2} \quad (145)$$

$$\alpha = \left( \frac{27\sqrt{3}}{8\pi^2} \right)^{2/5} \cdot \xi^{11/5} \quad (146)$$

### Numerical Values

23.5.1 Step-by-step calculation:

$$\frac{27\sqrt{3}}{8\pi^2} \approx \frac{46.765}{78.956} \approx 0.5923 \quad (147)$$

$$\left( \frac{27\sqrt{3}}{8\pi^2} \right)^{2/5} \approx (0.5923)^{0.4} \approx 0.8327 \quad (148)$$

$$\xi^{11/5} = \xi^{2.2} = \left( \frac{4}{3} \times 10^{-4} \right)^{2.2} \quad (149)$$

**With**  $\xi = 4/3 \times 10^{-4}$

### 23.6.1 Final calculation:

$$\xi = 1.333333 \times 10^{-4} \quad (150)$$

$$\xi^{2.2} \approx (1.333333 \times 10^{-4})^{2.2} \quad (151)$$

$$\approx 8.758 \times 10^{-9} \quad (152)$$

$$\alpha \approx 0.8327 \times 8.758 \times 10^{-9} \quad (153)$$

$$\approx 7.292 \times 10^{-3} \quad (154)$$

$$\alpha^{-1} \approx 137.13 \quad (155)$$

## Symbol Explanation

### 23.7.1 Key symbols used:

|            |                                                             |
|------------|-------------------------------------------------------------|
| $\alpha$   | Fine structure constant ( $\approx 1/137.036$ )             |
| $\xi$      | Geometric space constant ( $= \frac{4}{3} \times 10^{-4}$ ) |
| $c_e$      | Electron mass coefficient                                   |
| $c_\mu$    | Muon mass coefficient                                       |
| $\pi$      | Pi ( $\approx 3.14159$ )                                    |
| $\sqrt{3}$ | Square root of 3 ( $\approx 1.73205$ )                      |
| $m_e$      | Electron mass ( $= 0.5109989461$ MeV)                       |
| $m_\mu$    | Muon mass ( $= 105.6583745$ MeV)                            |

## With Fractal Correction

### 23.8.1 Including the fractal factor:

$$\alpha^{-1} = \frac{7500}{m_e m_\mu} \cdot \left(1 - \frac{D_f - 2}{68}\right) = 138.949 \times 0.9862 = 137.036$$

## Final Fundamental Relationship

### 23.9.1 The complete formula:

$$\alpha = \left(\frac{27\sqrt{3}}{8\pi^2}\right)^{2/5} \cdot \xi^{11/5} \cdot K_{\text{frac}} \quad \text{with} \quad K_{\text{frac}} = 0.9862$$

## .24 Observation: $\alpha$ Cancels Out

### Equating the Formula Sets

#### 24.1.1 Comparing two representations:

$$\text{Simple: } m_e = \frac{2}{3} \cdot \xi^{5/2}$$

$$\text{T0-Theory: } m_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \xi^{5/2}$$

After dividing by  $\xi^{5/2}$ :

$$\frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}$$

## Solving for $\alpha$

### 24.2.1 Algebraic solution:

$$\alpha^{1/2} = \frac{3\sqrt{3}}{2\pi} \cdot \frac{3}{2} = \frac{9\sqrt{3}}{4\pi} \Rightarrow \alpha = \left( \frac{9\sqrt{3}}{4\pi} \right)^2 = \frac{243}{16\pi^2}$$

## For the Muon

### 24.3.1 Similar analysis:

$$\text{Simple: } m_\mu = \frac{8}{5} \cdot \xi^2$$

$$\text{T0-Theory: } m_\mu = \frac{9}{4\pi\alpha} \cdot \xi^2$$

After dividing by  $\xi^2$ :

$$\frac{8}{5} = \frac{9}{4\pi\alpha} \Rightarrow \alpha = \frac{9}{4\pi} \cdot \frac{5}{8} = \frac{45}{32\pi}$$

## The Apparent Contradiction

### 24.4.1 Three different values:

$$\text{From electron: } \alpha = \frac{243}{16\pi^2} \approx 1.539$$

$$\text{From muon: } \alpha = \frac{45}{32\pi} \approx 0.4474$$

$$\text{Experimental: } \alpha \approx 0.007297$$

## Resolution in the T0 Framework

### 24.5.1 In the T0 framework, $\alpha$ is not a free parameter but a function of $\xi$ :

$$\begin{array}{l} \frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \\ \frac{8}{5} = \frac{9}{4\pi\alpha} \end{array} \Rightarrow \alpha = \alpha(\xi)$$

## The Key Elements

### 24.6.1 The key elements:

1. The **geometric factors** ( $3\sqrt{3}/2\pi$ ,  $9/4\pi$ )
  2. The **powers of  $\alpha$**  ( $\alpha^{-1/2}$ ,  $\alpha^{-1}$ )
  3. The **rational coefficients** ( $2/3$ ,  $8/5$ )
- are set up so that they **compensate** each other.

## Meaning of the Different Representations

### 24.7.1 Comparative analysis:

- **Simple formulas:**  $m_e = \frac{2}{3}\xi^{5/2}$ ,  $m_\mu = \frac{8}{5}\xi^2$ 
  - Show the pure  $\xi$ -dependence

- Compact and transparent
- **Extended formulas:**  $m_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}\xi^{5/2}$ ,  $m_\mu = \frac{9}{4\pi\alpha}\xi^2$ 
  - Show the **origin** of the coefficients
  - Connect geometry ( $\pi, \sqrt{3}$ ) with EM coupling ( $\alpha$ )
  - But:  $\alpha$  is thereby **fixed**, not freely choosable

## Central Statement

### 24.8.1 The central insight:

In the T0 framework the lepton mass hierarchy is fixed by  $\xi$

The different mathematical representations are equivalent descriptions of the same fundamental geometry.

## Why This Insight Is Important

### 24.9.1 The implications:

1. **Unity:** The lepton masses follow from the one parameter  $\xi$  (plus structural choices as settings)
2. **Geometric basis:** The coefficients stem from fundamental geometry
3.  $\alpha$  **is derived:** The fine structure constant appears as a secondary quantity
4. **Simple structure:** The formulas are compact; mathematical simplicity is a hint, not evidence

## .25 Why the Extended Form Is Important

### The Two Equivalent Representations

#### 25.1.1 Comparing formulations:

**Simple form:**  $m_e = \frac{2}{3} \cdot \xi^{5/2}$

**Extended form:**  $m_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \xi^{5/2}$

### The Apparent Contradiction

#### 25.2.1 When equating both formulas:

$$\frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}$$

This yields for  $\alpha$ :

$$\alpha = \left( \frac{9\sqrt{3}}{4\pi} \right)^2 = \frac{243}{16\pi^2} \approx 1.539$$

## Consequence

### 25.3.1 The fractions cannot simply cancel out.

The extended form shows that the apparently simple fraction  $\frac{2}{3}$  is actually composed of more fundamental geometric and physical constants:

$$\frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}$$

## Mathematical Structure

### 25.4.1 The decomposition:

$$\frac{2}{3} = \frac{\text{Geometry factor}}{\alpha^{1/2}}$$

$$\text{with Geometry factor} = \frac{3\sqrt{3}}{2\pi} \approx 0.826$$

## Physical Interpretation

### 25.5.1 Interpretation:

- $\frac{2}{3}$  is **not** a simple rational fraction
- In the extended form it is composed of:
  - Space geometry ( $\pi, \sqrt{3}$ )
  - Electromagnetic coupling ( $\alpha$ )
  - Quantum numbers (implicit in the coefficients)
- The extended form shows this origin

## Why Both Representations Are Important

### 25.6.1 Complementary perspectives:

#### Simple Form

Shows pure  $\xi$ -dependence  
Compact  
Practical for calculations  
Hides the composition

#### Extended Form

Shows physical origin  
Shows the physical origin  
Fundamental for understanding  
Shows the composition

## The Actual Statement of T0-Theory

### 25.7.1 Core statement:

$$\frac{2}{3} \neq \text{simple fraction} \quad \text{but rather} \quad \frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}$$

**The extended form is necessary to show:**

1. That the fractions do **not** simply cancel
2. That the apparently simple coefficient  $\frac{2}{3}$  actually has a complex structure
3. That  $\alpha$  is part of this structure, even if it formally cancels out
4. That the geometry of space  $(\pi, \sqrt{3})$  is embedded in it

## Why No Fractal Correction is Needed for Mass Ratios and Characteristic Energy

### 1. Different Calculation Approaches

**Path A:**  $\alpha = \frac{m_e m_\mu}{7500}$  (requires correction)

**Path B:**  $\alpha = \frac{E_0^2}{7500}$  (requires correction)

**Path C:**  $\frac{m_\mu}{m_e} = f(\alpha)$  (no correction needed)

**Path D:**  $E_0 = \sqrt{m_e m_\mu}$  (no correction needed)

### 2. Mass Ratios Are Correction-Free

The lepton mass ratio:

$$\frac{m_\mu}{m_e} = \frac{c_\mu \xi^2}{c_e \xi^{5/2}} = \frac{c_\mu}{c_e} \xi^{-1/2}$$

Substituting the coefficients:

$$\frac{m_\mu}{m_e} = \frac{\frac{9}{4\pi\alpha}}{\frac{3\sqrt{3}}{2\pi\alpha^{1/2}}} \cdot \xi^{-1/2} = \frac{3\sqrt{3}}{2\alpha^{1/2}} \cdot \xi^{-1/2}$$

### 3. Why the Ratio is Correct

**The fractal correction cancels out in the ratio.**

$$\frac{m_\mu}{m_e} = \frac{K_{\text{frac}} \cdot m_\mu}{K_{\text{frac}} \cdot m_e} = \frac{m_\mu}{m_e}$$

The same correction factor affects both masses and cancels in the ratio.

### 4. Characteristic Energy is Correction-Free

$$E_0 = \sqrt{m_e m_\mu} = \sqrt{K_{\text{frac}} m_e \cdot K_{\text{frac}} m_\mu} = K_{\text{frac}} \cdot \sqrt{m_e m_\mu}$$

However:  $E_0$  is itself an observable. The corrected characteristic energy is:

$$E_0^{\text{corr}} = \sqrt{m_e^{\text{corr}} m_\mu^{\text{corr}}} = K_{\text{frac}} \cdot E_0^{\text{bare}}$$

## 5. Consistent Treatment

$$m_e^{\text{exp}} = K_{\text{frac}} \cdot m_e^{\text{bare}}$$

$$m_\mu^{\text{exp}} = K_{\text{frac}} \cdot m_\mu^{\text{bare}}$$

$$E_0^{\text{exp}} = K_{\text{frac}} \cdot E_0^{\text{bare}}$$

## 6. Calculating $\alpha$ via Mass Ratio

$$\frac{m_\mu}{m_e} = \frac{105.6583745}{0.5109989461} = 206.768282$$

Theoretical prediction (without correction):

$$\frac{m_\mu}{m_e} = \frac{8/5}{2/3} \cdot \xi^{-1/2} = \frac{12}{5} \cdot \xi^{-1/2}$$

## 7. Why Different Paths Require Different Treatments

| No Correction Needed        | Correction Required              |
|-----------------------------|----------------------------------|
| Mass ratios                 | Absolute mass values             |
| Characteristic energy $E_0$ | Fine structure constant $\alpha$ |
| Scale ratios                | Absolute energies                |
| Dimensionless quantities    | Dimensionful quantities          |

## 8. Physical Interpretation

- **Relative quantities:** Ratios are independent of absolute scale
- **Absolute quantities:** Require correction for absolute energy scale
- **Fractal dimension:** Affects absolute scaling, not ratios

## 9. Mathematical Reason

The fractal correction acts as a multiplicative factor:

$$m^{\text{exp}} = K_{\text{frac}} \cdot m^{\text{bare}}$$

For ratios:

$$\frac{m_1^{\text{exp}}}{m_2^{\text{exp}}} = \frac{K_{\text{frac}} \cdot m_1^{\text{bare}}}{K_{\text{frac}} \cdot m_2^{\text{bare}}} = \frac{m_1^{\text{bare}}}{m_2^{\text{bare}}}$$

## 10. Comparison with Experiment

$$\left(\frac{m_\mu}{m_e}\right)_{\text{exp}} = 206.768282$$

$$\left(\frac{m_\mu}{m_e}\right)_{\text{theo}} = 206.768282 \quad (\text{without correction})$$

# Is This Indirect Proof That the Fractal Correction is Correct?

## The Consistency Argument

The behaviour is consistent with the fractal correction. Since a common multiplicative factor cancels in every ratio, this is a consistency hint, but on its own not yet independent evidence.

## 1. The Theoretical Framework

The T0-theory proposes:

$$\begin{aligned} m_e &= \frac{2}{3} \cdot \xi^{5/2} \cdot K_{\text{frac}} \\ m_\mu &= \frac{8}{5} \cdot \xi^2 \cdot K_{\text{frac}} \\ \alpha &= \frac{m_e m_\mu}{7500} \cdot \frac{1}{K_{\text{frac}}} \end{aligned}$$

## 2. The Consistency Test

If the fractal correction is valid, then:

$$\frac{m_\mu}{m_e} = \frac{\frac{8}{5} \cdot \xi^2 \cdot K_{\text{frac}}}{\frac{2}{3} \cdot \xi^{5/2} \cdot K_{\text{frac}}} = \frac{12}{5} \cdot \xi^{-1/2}$$

## 3. Experimental Verification

$$\begin{aligned} \left(\frac{m_\mu}{m_e}\right)_{\text{theo}} &= \frac{12}{5} \cdot (1.333 \times 10^{-4})^{-1/2} \\ &= 2.4 \times 86.6 = 207.84 \\ \left(\frac{m_\mu}{m_e}\right)_{\text{exp}} &= 206.768 \end{aligned}$$

The 0.5% difference is not broken down further in this document.

## 4. What Speaks for the Correction

1. **Self-consistency:** The correction cancels where the multiplicative ansatz leads one to expect it
2. **Predictive power:** Mass ratios work without correction
3. **Explanatory power:** Absolute values need correction
4. **Parameter economy:** One correction factor ( $K_{\text{frac}}$ ) describes the deviations of the absolute values considered here

## 5. Comparison with Alternative Theories

Without fractal correction:

$$\begin{aligned}\alpha^{-1} &= 138.93 \quad (\text{calculated}) \\ \alpha^{-1} &= 137.036 \quad (\text{experimental}) \\ \text{Error} &= 1.38\%\end{aligned}$$

With fractal correction:

$$\alpha^{-1} = 138.93 \times 0.9862 = 137.036 \quad (\text{with correction})$$

## 6. The Philosophical Argument

**That the correction is needed for absolute values and cancels in ratios is compatible with its interpretation as a real physical effect. Whether it is more than a fitting factor will be decided by an independent derivation of  $K_{\text{frac}}$ .**

## 7. Further Connections

- The correction factor  $K_{\text{frac}} = 0.9862$  is motivated in the T0 framework by fractal geometry
- It connects to the effective fractal dimension  $D_f^{\text{eff}} \approx 2.973$  of spacetime
- For the value  $C = 68$  a connection to tetrahedral symmetry is conjectured **[S]**

## 9. Remaining Open Questions

- Direct measurement of spacetime's fractal dimension
- Extension to other particle families